```
import requests
import json
```

```
data = requests.get("https://web.archive.org/web/20240929211114/https://fruityvice.com/api/fruit/all")
data.raise_for_status()
data.headers.get("content-type")
data.text
```


```
'[{"name":"Persimmon","id":52,"family":"Ebenaceae","order":"Rosales","genus":"Diospyros","nutritions":{"calories":81,"fat":0.0,"sugar":18.0,"carbohydrates":18.0,"protein":0.0}},{"name":"Strawberry","id":3,"family":"Rosaceae","order":"Rosales","genus":"Fragaria","nutritions":{"calories":29,"fat":0.4,"sugar":5.4,"carbohydrates":5.5,"protein":0.8}},{"name":"Banana","id":1,"family":"Musaceae","order":"Zingiberales","genus":"Musa","nutritions":{"calories":96,"fat":0.2,"sugar":17.2,"carbohydrates":22.0,"protein":1.0}},{"name":"Tomato","id":5,"family":"Solanaceae","order":"Solanales","genus":"Solanum","nutritions":{"calories":74,"fat":0.2,"sugar":2.6,"carbohydrates":3.9,"protein":0.9}},{"name":"Pear","id":4,"family":"Rosaceae","order":"Rosales","genus":"Pyrus","nutritions":{"calories":57,"fat":0.1,"sugar":10.0,"carbohydrates":15.0,"protein":0.4}},{"name":"Durian","id":60,"family":"Malvaceae","order":"Malvales","genus":"Durio","nutritions":{"calories":147,"fat":5.3,"sugar":6.75,"carbohydrates":…'
```


```
results = json.loads(data.text)
```

```
pd.DataFrame(results).head()
```

	name	id	family	order	genus	nutritions
0	Persimmon	52	Ebenaceae	Rosales	Diospyros	{'calories': 81, 'fat': 0.0, 'sugar': 18.0, 'c...
1	Strawberry	3	Rosaceae	Rosales	Fragaria	{'calories': 29, 'fat': 0.4, 'sugar': 5.4, 'ca...
2	Banana	1	Musaceae	Zingiberales	Musa	{'calories': 96, 'fat': 0.2, 'sugar': 17.2, 'c...
3	Tomato	5	Solanaceae	Solanales	Solanum	{'calories': 74, 'fat': 0.2, 'sugar': 2.6, 'ca...
4	Pear	4	Rosaceae	Rosales	Pyrus	{'calories': 57, 'fat': 0.1, 'sugar': 10.0, 'c...

```
df2 = pd.json_normalize(results)
df2.head()
```

	name	id	family	order	genus	nutritions.calories	nutritions.fat	nutritions.sugar	nutritions.carbohydrates	nutritions.protein
0	Persimmon	52	Ebenaceae	Rosales	Diospyros	81	0.0	18.0	18.0	0.0
1	Strawberry	3	Rosaceae	Rosales	Fragaria	29	0.4	5.4	5.5	0.8
2	Banana	1	Musaceae	Zingiberales	Musa	96	0.2	17.2	22.0	1.0
3	Tomato	5	Solanaceae	Solanales	Solanum	74	0.2	2.6	3.9	0.9
4	Pear	4	Rosaceae	Rosales	Pyrus	57	0.1	10.0	15.0	0.4

```
cherry = df2.loc[df2["name"] == 'Cherry']
print((cherry.iloc[0]['family']) , (cherry.iloc[0]['genus']))
cherry.head()
```

```
Rosaceae Prunus
```

	name	id	family	order	genus	nutritions.calories	nutritions.fat	nutritions.sugar	nutritions.carbohydrates	nutritions.protein
37	Cherry	9	Rosaceae	Rosales	Prunus	50	0.3	8.0	12.0	1.0

```
cal_banana = df2.loc[df2["name"] == 'Banana']
cal_banana.iloc[0]['nutritions.calories']
```

```
np.int64(96)
```

```
data2 = requests.get("https://official-joke-api.appspot.com/jokes/ten")
```

```
data2.text
```


```
'[{"type":"knock-knock","setup":"Knock knock. \\n Who\'s there? \\n Cows go. \\n Cows go who?","punchline":"No, cows go moo.","id":12},{"type":"general","setup":"What did the dog say to the two trees?","punchline":"Bark bark.","id":170},{"type":"general","setup":"Have you heard of the band 1023MB?","punchline":"They haven\'t got a gig yet.","id":110},{"type":"general","setup":"What do you call a group of disorganized cats?","punchline":"A cat-tastrophe.","id":208},{"type":"general","setup":"What do you call a pig with three eyes?","punchline":"Piiig","id":213},{"type":"general","setup":"What are the strongest days of the week?","punchline":"Saturday and Sunday...the rest are weekdays.","id":154},{"type":"programming","setup":"Why did the programmer bring a broom to work?","punchline":"To clean up all the bugs.","id":446},{"type":"general","setup":"Why did the golfer bring two pairs of pants?","punchline":"In case he got a hole in one.","id":427},{"type":"general","setup":"How many bones…'
```


```
results2 = json.loads(data2.text)
```

```
df3 = pd.DataFrame(results2)
```

[ ]

```
df3
```

	type	setup	punchline	id
0	general	Why can't you hear a pterodactyl go to the bat...	The p is silent.	309
1	programming	Why was the font always tired?	It was always bold.	443
2	general	Why do pirates not know the alphabet?	They always get stuck at "C".	349
3	general	Where do sheep go to get their hair cut?	The baa-baa shop.	287
4	general	What did the shy pebble wish for?	That she was a little boulder.	184
5	general	A grocery store cashier asked if I would like ...	I told her 'No, thanks. The carton works fine.'	397
6	general	What do you call a dictionary on drugs?	High definition.	199
7	general	Why did the half blind man fall in the well?	Because he couldn't see that well!	325
8	general	What does a pirate pay for his corn?	A buccaneer!	235
9	general	Why do mathematicians hate the U.S.?	Because it's indivisible.	348

[ ]

```
df3.drop(columns=["type","id"],inplace=True)
```

[ ]

```
df3
```

	setup	punchline
0	Why can't you hear a pterodactyl go to the bat...	The p is silent.
1	Why was the font always tired?	It was always bold.
2	Why do pirates not know the alphabet?	They always get stuck at "C".
3	Where do sheep go to get their hair cut?	The baa-baa shop.
4	What did the shy pebble wish for?	That she was a little boulder.
5	A grocery store cashier asked if I would like ...	I told her 'No, thanks. The carton works fine.'
6	What do you call a dictionary on drugs?	High definition.
7	Why did the half blind man fall in the well?	Because he couldn't see that well!
8	What does a pirate pay for his corn?	A buccaneer!
9	Why do mathematicians hate the U.S.?	Because it's indivisible.

[ ]

```
!pip install pandas numpy
!pip install lxml
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas) (2026.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: lxml in /usr/local/lib/python3.13/dist-packages (6.1.2)
```

[ ]

```
import numpy as np
import pandas as pd
```

[ ]

```
URL="https://web.archive.org/web/20230902185326/https://en.wikipedia.org/wiki/List_of_countries_by_GDP_%28nominal%29"
```

```
tables = pd.read_html(URL)
df = tables[3]
df.head()
```

	Country/Territory	UN region	IMF[1][13]		World Bank[14]		United Nations[15]	
	Country/Territory	UN region	Estimate	Year	Estimate	Year	Estimate	Year
0	World	—	105568776	2023	100562011	2022	96698005	2021
1	United States	Americas	26854599	2023	25462700	2022	23315081	2021
2	China	Asia	19373586	[n 1]2023	17963171	[n 3]2022	17734131	[n 1]2021
3	Japan	Asia	4409738	2023	4231141	2022	4940878	2021
4	Germany	Europe	4308854	2023	4072192	2022	4259935	2021

```
df.columns = range(df.shape[1])
df.head()
```

	0	1	2	3	4	5	6	7
0	World	—	105568776	2023	100562011	2022	96698005	2021
1	United States	Americas	26854599	2023	25462700	2022	23315081	2021
2	China	Asia	19373586	[n 1]2023	17963171	[n 3]2022	17734131	[n 1]2021
3	Japan	Asia	4409738	2023	4231141	2022	4940878	2021
4	Germany	Europe	4308854	2023	4072192	2022	4259935	2021

```
df = df[[0,2]]
df.head()
```

	0	2
0	World	105568776
1	United States	26854599
2	China	19373586
3	Japan	4409738
4	Germany	4308854

[ ]

```
df = df.iloc[1:11,:]
df
```

	0	2
2	China	19373586
3	Japan	4409738
4	Germany	4308854
5	India	3736882
6	United Kingdom	3158938
7	France	2923489
8	Italy	2169745
9	Canada	2089672
10	Brazil	2081235

[ ]

```
df.columns = ['Country','GDP (Million USD)']
df
```

...

	Country	GDP (Million USD)
2	China	19373586
3	Japan	4409738
4	Germany	4308854
5	India	3736882
6	United Kingdom	3158938
7	France	2923489
8	Italy	2169745
9	Canada	2089672
10	Brazil	2081235

[ ]

```
df['GDP (Million USD)'] = df['GDP (Million USD)'].astype(int)
df
```

	Country	GDP (Million USD)
2	China	19373586
3	Japan	4409738
4	Germany	4308854
5	India	3736882
6	United Kingdom	3158938
7	France	2923489
8	Italy	2169745
9	Canada	2089672
10	Brazil	2081235

[ ]

```
df[['GDP (Million USD)']] = df[['GDP (Million USD)']]/1000
df
```

	Country	GDP (Million USD)
2	China	19.373586
3	Japan	4.409738
4	Germany	4.308854
5	India	3.736882
6	United Kingdom	3.158938
7	France	2.923489
8	Italy	2.169745
9	Canada	2.089672
10	Brazil	2.081235

```
df[['GDP (Million USD)']] = np.round(df[['GDP (Million USD)']], 2)
df
```

	Country	GDP (Million USD)
2	China	19.37
3	Japan	4.41
4	Germany	4.31
5	India	3.74
6	United Kingdom	3.16
7	France	2.92
8	Italy	2.17
9	Canada	2.09
10	Brazil	2.08

```
df.rename(columns = {'GDP (Million USD)' : 'GDP (Billion USD)'})
```

	Country	GDP (Billion USD)
2	China	19.37
3	Japan	4.41
4	Germany	4.31
5	India	3.74
6	United Kingdom	3.16
7	France	2.92
8	Italy	2.17
9	Canada	2.09
10	Brazil	2.08

Load the DataFrame to the CSV file named "Largest_economies.csv"

```
df.to_csv('./Largest_economies.csv')
```